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Autonomous SOCP–MPC for space debris pre-capture rendezvous and pointing

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Abstract

In autonomous active debris removal, defunct and uncontrolled satellites may retain residual maneuvering capability, turning the pre capture phase into a pursuit-evasion rendezvous that must maintain a safe flight corridor and continuous payload pointing for reliable capture readiness. This paper proposes an autonomous model predictive control (MPC) framework formulated as a second order cone program (SOCP), resulting in a pre capture SOCP–MPC that unifies orbital and attitude dynamics under actuator saturation and mission constraints. Flight corridor and pointing requirements are encoded as second order cone constraints with time varying half angles for dynamic constraint tightening, enabling early corridor entry and robust pointing without precomputed reference trajectories. Performance is evaluated against differential games and proportional derivative (DG–PD) control baselines with and without manual guidance. Results indicate earlier corridor entry, faster pointing alignment, and lower propellant usage, and reveal a tunable trade off between overall duration and cone entry timing.
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Keywords: Model predictive control; Second-order cone program; Pursuit-evasion games; Rendezvous and proximity operations; On orbit servicing; Active debris removal

1. Introduction

Space debris, especially large uncontrolled objects such as spent rocket bodies and retired satellites, has become an escalating threat to on orbit assets (Filho et al., 2025). In the geostationary orbit (GEO) belt, where natural decay is negligible, more than 1300 inactive satellites compete for limited station-keeping slots (ESA Space Debris Office, 2025). To mitigate these risks, active debris removal (ADR) concepts deploy a service spacecraft to autonomously approach, capture, and transfer the target to a disposal orbit. This work focuses on autonomous pursuit-evasion rendezvous and sustained attitude pointing against uncontrolled target debris.

Representative tether-based capture technologies (e.g., robotic-tether modules, tether nets, and tether harpoons) have been explored (Huang et al., 2017; Santos et al., 2022; Poozhiyil et al., 2023; Sharf et al., 2017; Aglietti et al., 2020; Forshaw et al., 2016). Tether-net capture is attractive because it can envelop complex shapes and does not require preliminary target stabilization (Lian et al., 2025). However, regardless of the capture method, precise

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attitude pointing and relative position keeping within a narrow flight corridor are necessary for providing a viable capture window (Aslanov and Ledkov, 2023). The overall ADR pre-capture mission can be framed as a pursuit-evasion rendezvous with continuous pointing.

Unlike conventional orbital pursuit-evasion games, formation keeping, and cooperative rendezvous, pursuit-evasion rendezvous for ADR pre-capture combines hard mission constraints, persistently evasive target motion, and critical observation conditions. During this phase, the service satellite must remain within a prescribed flight corridor to ensure observability. In parallel, pointing accuracy requires the payload direction to remain stably aligned with the target's center of mass (CM).

Related problems in orbital pursuit-evasion (OPE) and autonomous rendezvous have reported effective uses of differential games (DG) and model predictive control (MPC). DG, originated by Isaacs (1965), optimizes against intelligent adversaries under worst case criteria (Weintraub et al., 2020) and use saddle point strategies to describe equilibria in constrained dynamic conflicts (Pais and Leonard, 2010). Based on interacting multiple model estimation and smooth variable structure filters, Tang et al. (2021) proposed a switching guidance that blends a linear quadratic saddle point controller with a zero effort miss metric. Shi et al. (2023) formulated a game with J_2 perturbation and direction constrained thrust and solved a two-point boundary value problem to obtain an open loop saddle point. Liu et al. (2025) incorporated a model predictive framework into a simplified Stackelberg setting. Learning-based multi-agent methods such as deep deterministic policy gradient have also been explored for constrained impulsive OPE (Zhao et al., 2023).

MPC has been widely applied to autonomous rendezvous. Early constrained designs preserved convexity and real time solvability, for example quadratic programming (QP) with thrust and rate limits (Weiss et al., 2015). Subsequent efforts handled non-cooperative or tumbling targets through linearized collision avoidance constraints (Li et al., 2017) and progressed toward coupled orbit–attitude regulation for soft docking (Yang, 2019). Tube-based MPC with online uncertainty identification improved robustness under bounded disturbances (Oestreich et al., 2023), and the associated admissible sets admit formulations in quadratically constrained quadratic program (QCQP) and second order cone program (SOCP).

Convex planners for rendezvous encode line of sight approach corridors (Fear and Lightsey, 2024) and represent keep-in and keep-out geometry with conic sets (Ortolano and Geller, 2021). SOCP is efficient for attitude and actuator constraints in time-optimal planning (Xu et al., 2023), and sequential conic optimization produces feasible and convergent attitude trajectories under time-varying mass (Liu et al., 2024). At the solver level, ALTRO-C enforces second-order cone (SOC) constraints and achieves sub-millisecond MPC solves with strong warm starts (Jackson et al., 2021). For impulsive guidance with complex constraints, waypoint convex reachable set methods provide fuel-optimal maneuver sequences with provable bounds (Hunter and D'Amico, 2025). Within autonomous rendezvous, line of sight corridor and thrust limits have been encoded in convex QP for elliptical orbits (Li and Zhu, 2018), while fully coupled six degrees-of-freedom (6-DOF) nonlinear MPC enforces line of sight cones together with gimbaled thruster limits (Malladi et al., 2019).

Despite these advances, many formulations still treat the flight corridor and the payload pointing as soft targets or only partially enforced constraints. A unified 6-DOF scheme that autonomously guarantees corridor adherence, continuous pointing, and thrust magnitude limits is still missing, especially when the target performs persistent maneuvers in the OPE scenarios.

To bridge the gaps, we present a unified 6-DOF MPC framework based on SOCP. The emphasis of this work is on a control-oriented modeling and integration of specific mission constraints into an SOCP–MPC scheme, rather than on developing new numerical algorithms for solving SOCPs. The method recasts the quadratic MPC cost as an SOC constraint and introduces two other types of mission SOC constraints–flight corridor and pointing–as time-varying SOC constraints, which together yield a single conic program that is solvable onboard in real time using standard conic optimization solvers. The controller does not rely on pre-planned reference trajectories and directly outputs thrust and torque commands to keep the service satellite aligned with the corridor while continuously pointing at the target. A dynamic cone-tightening strategy is incorporated to account for persistent target maneuvers while preserving convexity on the service satellite side. In the case studies, the proposed method maintains the flight corridor and pointing cone more effectively than the decoupled differential games and proportional derivative (DG–PD) baselines, while also achieving lower propellant consumption and nearly optimal mission duration, thereby delivering superior overall performance.

2. Dynamics modeling

In the satellite pursuit-evasion rendezvous and pointing scenarios, the service satellite is capable of performing orbital and attitude maneuvers to approach the target from long range and is expected to maintain relative station within a prescribed close distance. Its attitude control ensures that the payload onboard the spacecraft remains continuously aligned with the target. In contrast, the target executes orbital maneuvers while remaining three-axis stabilized throughout the process, without conducting any additional attitude maneuvers.

For the purpose of modeling the orbital motion of the service and target satellites, a virtual reference satellite is introduced, which coincides with the target's position at the initial instant.

Assumption 1. The service satellite, the target, and the reference satellite are assumed to operate in GEO circular orbits. The reference satellite performs no orbital maneuvers and follows pure Keplerian motion throughout.

Fig. 1 illustrates the coordinate frames used in this paper, which are defined as follows:

(i) Geocentric equatorial inertial frame $\{O_iX_iY_iZ_i\}$: The origin O_i is located at the Earth's CM. The O_iX_i-axis points toward the vernal equinox, the O_iZ_i-axis is aligned with the Earth's rotation axis directed to the North Pole, and the O_iY_i-axis lies in the equatorial plane, determined by the right-hand rule.
(ii) Orbital frame $\{O_oX_oY_oZ_o\}$: The origin Oo is located at the CM of the reference satellite. The O_oZ_o-axis points toward the Earth, the O_oX_o-axis is aligned with the velocity direction of the reference orbit, and the O_oY_o-axis is perpendicular to the orbital plane, completing a right-handed triad.
(iii) Body-fixed frames of the service satellite $\{O_mX_mY_mZ_m\}$ and the target $\{O_tX_tY_tZ_t\}$: The origins are located at the respective spacecraft CMs, and the axes are rigidly attached to the spacecraft and aligned with their principal axes of inertia.
(iv) Pointing frame $\{O_pX_pY_pZ_p\}$: The origin O_p is located at the service satellite's CM, the O_pX_p-axis points from the target to the service satellite. The O_pY_p-axis is defined to be perpendicular to both the orbital-frame O_oZ_o-axis and the O_pX_p-axis. The O_pZ_p-axis is then determined by the right-hand rule to complete the orthogonal coordinate frame.

According to Assumption 1, the orbital radius of both the service satellite and the target is much larger than the relative motion range over a short time span. Therefore, the orbital dynamics can be represented by the linear Clohessy-Wiltshire (CW) equations (Tang et al., 2021).

Assumption 2. The perturbing forces acting on the satellites are neglected, as is the effect of fuel consumption on the mass of both the service satellite and the target.

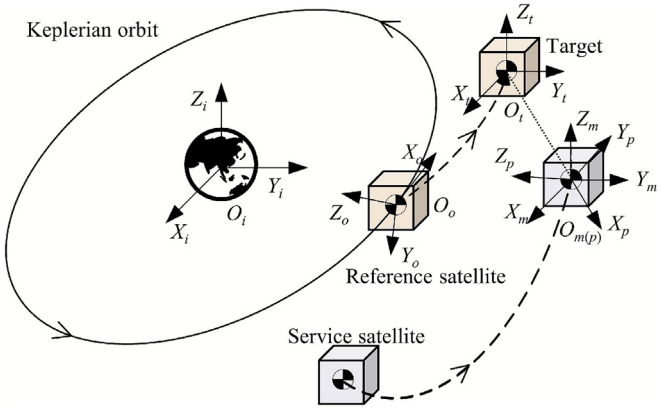


Fig. 1. Coordinate systems.

Under Assumption 2, the CW equations are expressed in the standard continuous-time state-space form, which is satisfied by both the service satellite and the target.

$$\dot{\boldsymbol{x}}_{mt} = \boldsymbol{A}_{pos}\boldsymbol{x}_{mt} + \boldsymbol{B}_{pos,\,m}\boldsymbol{F}_m + \boldsymbol{B}_{pos,\,t}\boldsymbol{F}_t \tag{1}$$

with the matrices,

$$\boldsymbol{x}_{mt} = \begin{bmatrix} \boldsymbol{r}_{mt}^T & \boldsymbol{v}_{mt}^T \end{bmatrix}^T, \quad \dot{\boldsymbol{x}}_{mt} = \begin{bmatrix} \dot{\boldsymbol{r}}_{mt}^T & \dot{\boldsymbol{v}}_{mt}^T \end{bmatrix}^T,$$

$$\boldsymbol{A}_{pos} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3\omega_o^2 & 0 & 0 & 0 & 2\omega_o & 0 \\ 0 & 0 & 0 & -2\omega_o & 0 & 0 \\ 0 & 0 & -\omega_o^2 & 0 & 0 & 0 \end{bmatrix} \triangleq \begin{bmatrix} \boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 \\ \boldsymbol{Q}_1 & \boldsymbol{Q}_2 \end{bmatrix},$$

$$\boldsymbol{B}_{pos,\,m} = \begin{bmatrix} \boldsymbol{0}_{3\times3} \\ m_m^{-1}\boldsymbol{I}_3 \end{bmatrix}, \quad \boldsymbol{B}_{pos,\,t} = \begin{bmatrix} \boldsymbol{0}_{3\times3} \\ -m_t^{-1}\boldsymbol{I}_3 \end{bmatrix},$$

where $\boldsymbol{x}_{mt}$ denotes the translational state vector of the service satellite relative to the target, consisting of the relative position and relative velocity $\boldsymbol{v}_{mt}$. $\boldsymbol{F}_m$ and $\boldsymbol{F}_t$ are the thrust vectors of the service satellite and the target, respectively. ω_o is the orbital angular velocity, $\boldsymbol{0}_{3\times3}$ denotes the 3×3 zero matrix, and $\boldsymbol{I}_3$ represents the third-order identity matrix. $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$ are coefficient matrices, while m_m and m_t are the masses of the service satellite and the target, respectively.

The attitude dynamics of both the service satellite and the target are described by the classical rigid-body Euler equations in angular velocity form (Xu et al., 2023).

$$\boldsymbol{J}_m\dot{\boldsymbol{\omega}}_m + \boldsymbol{\omega}_m \times (\boldsymbol{J}_m\boldsymbol{\omega}_m) = \boldsymbol{T}_m \tag{2}$$

$$\boldsymbol{J}_t\dot{\boldsymbol{\omega}}_t + \boldsymbol{\omega}_t \times (\boldsymbol{J}_t\boldsymbol{\omega}_t) = \boldsymbol{T}_t \tag{3}$$

where $\boldsymbol{J}_m$ and $\boldsymbol{J}_t$ denote the inertia matrices of the service satellite and the target, respectively. $\boldsymbol{\omega}_m$ and $\boldsymbol{\omega}_t$ are the angular velocity vectors of the service satellite and the target. $\boldsymbol{T}_m$ and $\boldsymbol{T}_t$ represent the external torque vectors acting on the service satellite and the target, respectively.

Define the quaternions of the service satellite and the target

$$\boldsymbol{Q}_m = q_{m0} + \boldsymbol{q}_m \tag{4}$$

$$\boldsymbol{Q}_t = q_{t0} + \boldsymbol{q}_t \tag{5}$$

where q_{m0}, q_{t0}, $\boldsymbol{q}_m$, $\boldsymbol{q}_t$ denote the scalar and vector parts of the corresponding quaternions.

The relative quaternion is defined as $\boldsymbol{Q}_{mt}$ which consists of the scalar part q_{mt0} and the vector part $\boldsymbol{q}_{mt}$. The associated operation is defined by the Kronecker product as follows:

$$\boldsymbol{Q}_{mt} = \boldsymbol{Q}_t^{-1} \otimes \boldsymbol{Q}_m = \begin{bmatrix} q_{mt0} \\ \boldsymbol{q}_{mt} \end{bmatrix} = \begin{bmatrix} q_{t0}q_{m0} + \boldsymbol{q}_t^T\boldsymbol{q}_m \\ q_{t0}\boldsymbol{q}_m - q_{m0}\boldsymbol{q}_t - \boldsymbol{q}_t^{\times}\boldsymbol{q}_m \end{bmatrix} \tag{6}$$

where the tilde sign "$\boldsymbol{a}^{\times}$" represents the antisymmetric matrix $\boldsymbol{a}^{\times}=\begin{bmatrix}0 & -a_3 & a_2\\ a_3 & 0 & -a_1\\ -a_2 & a_1 & 0\end{bmatrix}$ of the column vector $\boldsymbol{a}=\begin{bmatrix}a_1 & a_2 & a_3\end{bmatrix}^T$.

The relative angular velocity $\boldsymbol{\omega}_{mt}$ is defined as follows:

$$\boldsymbol{\omega}_{mt}=\boldsymbol{\omega}_m-\boldsymbol{C}_{mt}\boldsymbol{\omega}_t \tag{7}$$

In the equation, $\boldsymbol{C}_{mt}$ denotes the coordinate transformation matrix from the target body frame to the service satellite body frame.

Based on this, the expression for the relative angular acceleration can be derived as follows:

$$\begin{aligned}\dot{\boldsymbol{\omega}}_{mt}=&-\boldsymbol{J}_m^{-1}\boldsymbol{\omega}_{mt}^{\times}\boldsymbol{J}_m\boldsymbol{\omega}_{mt}-\boldsymbol{J}_m^{-1}(\boldsymbol{C}_{mt}\boldsymbol{\omega}_t)^{\times}\boldsymbol{J}_m\boldsymbol{\omega}_{mt}\\ &-\boldsymbol{J}_m^{-1}\boldsymbol{\omega}_{mt}^{\times}\boldsymbol{J}_m\boldsymbol{C}_{mt}\boldsymbol{\omega}_t-\boldsymbol{J}_m^{-1}(\boldsymbol{C}_{mt}\boldsymbol{\omega}_t)^{\times}\boldsymbol{J}_m\boldsymbol{C}_{mt}\boldsymbol{\omega}_t\\ &-\boldsymbol{\omega}_{mt}^{\times}\boldsymbol{C}_{mt}\boldsymbol{\omega}_t+\boldsymbol{C}_{mt}\boldsymbol{J}_t^{-1}\boldsymbol{\omega}_t^{\times}\boldsymbol{J}_t\boldsymbol{\omega}_t+\boldsymbol{J}_m^{-1}\boldsymbol{T}_m\\ &-\boldsymbol{C}_{mt}\boldsymbol{J}_t^{-1}\boldsymbol{T}_t\end{aligned} \tag{8}$$

Consequently, the attitude kinematic equations are formulated as follows:

$$\dot{\boldsymbol{q}}_{mt}=\frac{1}{2}(q_{mt0}\boldsymbol{I}_3+\boldsymbol{q}_{mt}^{\times})\boldsymbol{\omega}_{mt}\triangleq\boldsymbol{G}_1\boldsymbol{\omega}_{mt} \tag{9}$$

where $\boldsymbol{G}_1$ is the coefficient matrix.

The state-space vector is defined as

$$\boldsymbol{X}=\begin{bmatrix}\boldsymbol{r}_{mt}^T & \boldsymbol{v}_{mt}^T & \boldsymbol{q}_{mt}^T & \boldsymbol{\omega}_{mt}^T\end{bmatrix}^T \tag{10}$$

According to Eqs. (1), (8), and (9), the state-space model can be expressed as

$$\begin{aligned}\dot{\boldsymbol{X}}\triangleq&\boldsymbol{A}\boldsymbol{X}+\boldsymbol{B}_m\boldsymbol{U}_m+\boldsymbol{B}_t\boldsymbol{U}_t+\boldsymbol{f}\\ =&\begin{bmatrix}\boldsymbol{0}_{3\times3} & \boldsymbol{I}_3 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{Q}_1 & \boldsymbol{Q}_2 & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{G}_1\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3} & \boldsymbol{G}_2\end{bmatrix}\begin{bmatrix}\boldsymbol{r}_{mt}\\ \boldsymbol{v}_{mt}\\ \boldsymbol{q}_{mt}\\ \boldsymbol{\omega}_{mt}\end{bmatrix}+\begin{bmatrix}\boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3}\\ \boldsymbol{f}_{att}\end{bmatrix}\\ &+\begin{bmatrix}\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ m_m^{-1}\boldsymbol{I}_3 & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{J}_m^{-1}\end{bmatrix}\begin{bmatrix}\boldsymbol{F}_m\\ \boldsymbol{T}_m\end{bmatrix}+\begin{bmatrix}\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ -m_t^{-1}\boldsymbol{I}_3 & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & -\boldsymbol{C}_{mt}\boldsymbol{J}_t^{-1}\end{bmatrix}\begin{bmatrix}\boldsymbol{F}_t\\ \boldsymbol{T}_t\end{bmatrix}\end{aligned} \tag{11}$$

where the coefficient matrix

$$\begin{aligned}\boldsymbol{G}_2=&-\boldsymbol{J}_m^{-1}\boldsymbol{\omega}_{mt}^{\times}\boldsymbol{J}_m-\boldsymbol{J}_m^{-1}(\boldsymbol{C}_{mt}\boldsymbol{\omega}_t)^{\times}\boldsymbol{J}_m\\ &+\boldsymbol{J}_m^{-1}(\boldsymbol{J}_m\boldsymbol{C}_{mt}\boldsymbol{\omega}_t)^{\times}+(\boldsymbol{C}_{mt}\boldsymbol{\omega}_t)^{\times},\end{aligned}$$

and the nonlinear term

$$\boldsymbol{f}_{att}=\boldsymbol{C}_{mt}\boldsymbol{J}_t^{-1}\boldsymbol{\omega}_t^{\times}\boldsymbol{J}_t\boldsymbol{\omega}_t-\boldsymbol{J}_m^{-1}(\boldsymbol{C}_{mt}\boldsymbol{\omega}_t)^{\times}\boldsymbol{J}_m(\boldsymbol{C}_{mt}\boldsymbol{\omega}_t).$$

3. SOCP–MPC scheme

3.1. Discrete-time 6-DOF prediction model and MPC formulation

Based on the continuous 6-DOF model in Section 2, this subsection derives the discrete-time prediction model and the quadratic MPC formulation used for rendezvous and pointing.

To address the specific constraints of the rendezvous and pointing mission while ensuring both real-time feasibility and robustness of online computation, the SOCP–MPC framework is introduced. The quadratic optimization problem with conic constraints is reformulated into a standard SOCP convex optimization format. Furthermore, by online updating of SOC constraint parameters, the approach and pointing trajectory are dynamically tightened, enabling rapid entry into the flight corridor and early alignment without prescribing any reference trajectory guidance. The block diagram of the proposed control scheme is shown in Fig. 2.

Assumption 3. The target only performs orbital maneuvers, while its attitude remains three-axis stabilized. The target's relative position, velocity, attitude, control inputs are known in advance.

According to Assumption 3, the term $\boldsymbol{f}_{att}$ in the dynamic model can be neglected, and Eq. (11) is simplified as

$$\dot{\boldsymbol{X}}(t)=\boldsymbol{A}\boldsymbol{X}(t)+\boldsymbol{B}_m\boldsymbol{U}_m(t)+\boldsymbol{B}_t\boldsymbol{U}_t(t) \tag{12}$$

During each sampling interval $[k\Delta t,\ (k+1)\Delta t)$, the control input is assumed to be held constant using zero-order hold (ZOH), i.e.,

$$\boldsymbol{U}_m(t)=\boldsymbol{U}_m(k),\ \boldsymbol{U}_t(t)=\boldsymbol{U}_t(k),\ t\in[k\Delta t,\ (k+1)\Delta t) \tag{13}$$

Eq. (12) can then be discretized as

$$\boldsymbol{X}(k+1)=\boldsymbol{A}_d\boldsymbol{X}(k)+\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k)+\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k) \tag{14}$$

where

$$\boldsymbol{A}_d=e^{\boldsymbol{A}\Delta t},\quad \boldsymbol{B}_{d,\,m}=\int_0^{\Delta t}e^{\boldsymbol{A}s}\boldsymbol{B}_m ds,\quad \boldsymbol{B}_{d,\,t}=\int_0^{\Delta t}e^{\boldsymbol{A}s}\boldsymbol{B}_t ds$$

In numerical simulations, it is difficult to directly compute the above integral. Instead, the block-matrix exponential method can be employed to obtain the discrete matrices.

Within the MPC framework, at each sampling instant k, the controller reads the current system state $\boldsymbol{X}(k)$, and then performs receding-horizon prediction of the future state sequence $\boldsymbol{X}(k+i|k)$ $(i=1,\ \cdots,\ N)$ over a finite prediction horizon N.

In the design of the service satellite's MPC controller, it is approximated that the control inputs of the target remain constant within the prediction horizon, i.e.,

$$\boldsymbol{U}_t(k+i|k)=\boldsymbol{U}_t(k)\ (i=1,\ \cdots,\ N) \tag{15}$$

By recursively expanding Eq. (14), the future state sequence over a finite prediction horizon N is obtained as

$$\boldsymbol{X}(k+1|k)=\boldsymbol{A}_d\boldsymbol{X}(k)+\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k)+\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k) \tag{16}$$

$$\begin{aligned}\boldsymbol{X}(k+2|k)=&\boldsymbol{A}_d^2\boldsymbol{X}(k)+\boldsymbol{A}_d\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k)\\ &+\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k+1|k)\\ &+\boldsymbol{A}_d\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k)+\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k)\end{aligned} \tag{17}$$

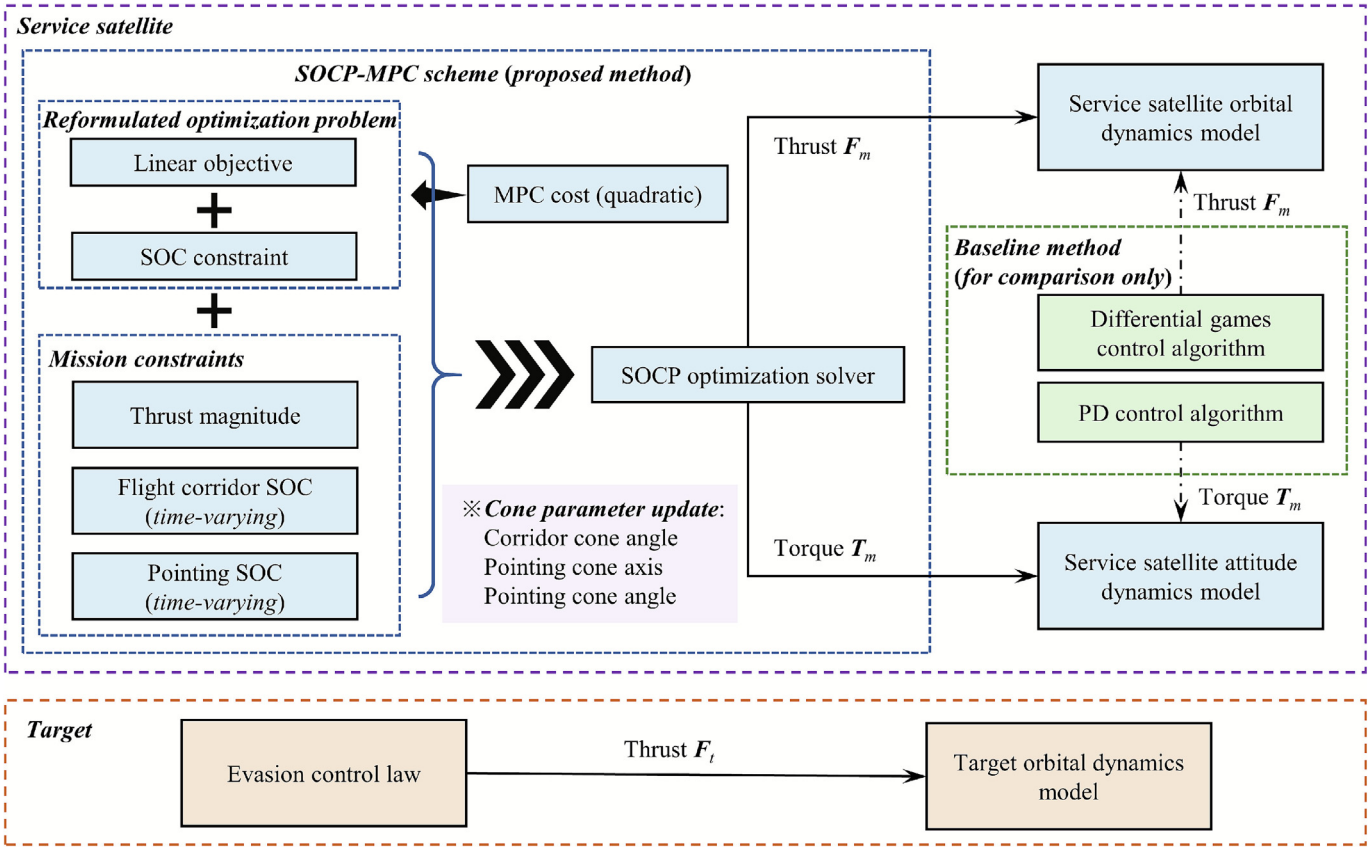


Fig. 2. SOCP–MPC architecture for OPE rendezvous and pointing.

$$
\begin{aligned}
&\cdots\cdots\\
&\boldsymbol{X}(k+N|k)=\boldsymbol{A}_d^N\boldsymbol{X}(k)\\
&\quad+\boldsymbol{A}_d^{N-1}\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k)\\
&\quad+\boldsymbol{A}_d^{N-2}\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k+1|k)\\
&+\cdots+\boldsymbol{B}_{d,\,m}\boldsymbol{U}_m(k+N-1|k)\\
&\quad+\boldsymbol{A}_d^{N-1}\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k)\\
&\quad+\boldsymbol{A}_d^{N-2}\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k)\\
&\quad+\cdots+\boldsymbol{B}_{d,\,t}\boldsymbol{U}_t(k)
\end{aligned}
\tag{18}
$$

By stacking all predicted states, let $\boldsymbol{U}_{m,\,k}$ denote the control input sequence within the prediction horizon, and $\boldsymbol{X}_k$ denote the corresponding state sequence.

$$
\boldsymbol{X}_k=\begin{bmatrix}\boldsymbol{X}(k)\\ \boldsymbol{X}(k+1|k)\\ \boldsymbol{X}(k+2|k)\\ \cdots\\ \boldsymbol{X}(k+N|k)\end{bmatrix}
\tag{19}
$$

$$
\boldsymbol{U}_{m,\,k}=\begin{bmatrix}\boldsymbol{U}_m(k)\\ \boldsymbol{U}_m(k+1|k)\\ \boldsymbol{U}_m(k+2|k)\\ \cdots\\ \boldsymbol{U}_m(k+N-1|k)\end{bmatrix}
\tag{20}
$$

The multi-step state prediction model given in Eqs. (16)–(18) can be combined and written as

$$
\boldsymbol{X}_k=\boldsymbol{M}\boldsymbol{X}(k)+\boldsymbol{N}\boldsymbol{U}_{m,\,k}+\boldsymbol{\Gamma}\boldsymbol{U}_t(k)
\tag{21}
$$

where the coefficient matrices are

$$
\boldsymbol{M}=\begin{bmatrix}\boldsymbol{I}\\ \boldsymbol{A}_d\\ \boldsymbol{A}_d^2\\ \cdots\\ \boldsymbol{A}_d^N\end{bmatrix},\quad
\boldsymbol{N}=\begin{bmatrix}\boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{0}\\ \boldsymbol{B}_{d,\,m} & \boldsymbol{0} & \cdots & \boldsymbol{0}\\ \boldsymbol{A}_d\boldsymbol{B}_{d,\,m} & \boldsymbol{B}_{d,\,m} & \cdots & \boldsymbol{0}\\ \vdots & \vdots & \ddots & \vdots\\ \boldsymbol{A}_d^{N-1}\boldsymbol{B}_{d,\,m} & \boldsymbol{A}_d^{N-2}\boldsymbol{B}_{d,\,m} & \cdots & \boldsymbol{B}_{d,\,m}\end{bmatrix},
$$

$$
\boldsymbol{\Gamma}=\begin{bmatrix}\boldsymbol{0}^T & \boldsymbol{B}_{d,\,t}^T & (\boldsymbol{A}_d\boldsymbol{B}_{d,\,t}+\boldsymbol{B}_{d,\,t})^T & \cdots & \left(\sum_{i=0}^{N-1}\boldsymbol{A}_d^i\boldsymbol{B}_{d,\,t}\right)^T\end{bmatrix}^T.
$$

Assume that the debris capture payload and measurement sensors are mounted along the X-axis of the service satellite body-fixed frame. As illustrated in Fig. 3, the service satellite must adjust its relative orbital position to enter the target's flight corridor cone as quickly as possible. The corridor deviation angle ϕ is defined as the angle between the relative position vector and the target's X-axis. The control objective is to drive the service satellite into the desired cone with half-angle ϕ_{des} at the earliest possible time. Similarly, the pointing deviation angle δ is defined as the angle between the relative position vector

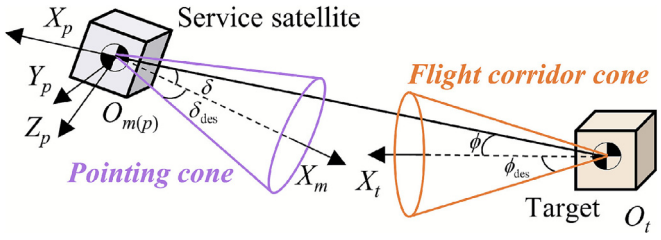


Fig. 3. Conic constraints.

and the service satellite's X-axis. By adjusting its attitude to reorient the cone axis, the service satellite aims to minimize the misalignment of the X-axis, such that the pointing error is reduced below the desired angle δ_{des} as quickly as possible.

The service satellite is prescribed to maintain a constant distance r_{ref} along the target's O_tX_t-axis. As a result, the reference relative position and velocity are uniquely determined by the target's state. Under Assumption 3, this state can be accurately estimated. The unit vectors of the pointing frame are then used to compute the corresponding reference attitude quaternion and reference angular velocity. By assembling the reference position, velocity, attitude quaternion, and angular velocity into the reference state $\boldsymbol{X}_{ref}$, the orbit–attitude maneuvering objective is achieved by driving the state vector $\boldsymbol{X}$ in Eq. (11) to track $\boldsymbol{X}_{ref}$.

Based on the state error sequence and the control input sequence, the complete receding horizon quadratic cost function is constructed as follows:

$$\begin{aligned} J_m = \sum_{i=0}^{N-1} \Big(& \big(\boldsymbol{X}(k+i|k) - \boldsymbol{X}_{ref}(k+i|k)\big)^T \\ & \boldsymbol{Q}\big(\boldsymbol{X}(k+i|k) - \boldsymbol{X}_{ref}(k+i|k)\big) \\ & + \boldsymbol{U}_m(k+i|k)^T \boldsymbol{R}_m \boldsymbol{U}_m(k+i|k) \\ & - \boldsymbol{U}_t(k)^T \boldsymbol{R}_t \boldsymbol{U}_t(k)\big) \\ & + \big(\boldsymbol{X}(k+N|k) - \boldsymbol{X}_{ref}(k+i|k)\big)^T \\ & \boldsymbol{F}\big(\boldsymbol{X}(k+N|k) - \boldsymbol{X}_{ref}(k+i|k)\big) \end{aligned} \tag{22}$$

where $\boldsymbol{Q}$, $\boldsymbol{R}_m$, $\boldsymbol{R}_t$, $\boldsymbol{F}$ are control parameter matrices.

Within a receding horizon, the small variations of the reference state over a short time are neglected, and the reference state sequence is defined as

$$\bar{\boldsymbol{X}}_{ref}(k) = \begin{bmatrix} \boldsymbol{X}_{ref}(k) \\ \boldsymbol{X}_{ref}(k) \\ \boldsymbol{X}_{ref}(k) \\ \vdots \\ \boldsymbol{X}_{ref}(k) \end{bmatrix} \tag{23}$$

The control parameter matrix is defined as

$$\bar{\boldsymbol{Q}} = \text{diag}(\boldsymbol{Q}, \cdots, \boldsymbol{Q}, \boldsymbol{F}) \tag{24}$$

$$\bar{\boldsymbol{R}} = \text{diag}(\boldsymbol{R}_m, \cdots, \boldsymbol{R}_m) \tag{25}$$

Here, the function diag$(\cdot)$ denotes either a diagonal matrix or a block-diagonal matrix, depending on the structure of the parameters.

According to Eq. (15), the terms associated with the target's control input can be approximated as constants. Therefore, Eq. (22) can be rewritten as

$$J_m \approx \left(\boldsymbol{X}_k - \bar{\boldsymbol{X}}_{ref}\right)^T \bar{\boldsymbol{Q}} \left(\boldsymbol{X}_k - \bar{\boldsymbol{X}}_{ref}\right) + \boldsymbol{U}_{m,\,k}^T \bar{\boldsymbol{R}}\, \boldsymbol{U}_{m,\,k} \tag{26}$$

By substituting Eq. (21) into Eq. (26) and expanding, while neglecting the constant terms in the cost function associated with the initial state $\boldsymbol{X}(k)$ and the reference state sequence $\bar{\boldsymbol{X}}_{ref}$, the cost function can be further simplified as

$$\begin{aligned} J_m = & \tfrac{1}{2} \boldsymbol{U}_{m,\,k}^T \left(\boldsymbol{N}^T \bar{\boldsymbol{Q}} \boldsymbol{N} + \bar{\boldsymbol{R}}\right) \boldsymbol{U}_{m,\,k} \\ & + \left(\boldsymbol{N}^T \bar{\boldsymbol{Q}} \left(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k) - \bar{\boldsymbol{X}}_{ref}\right)\right)^T \boldsymbol{U}_{m,\,k} \end{aligned} \tag{27}$$

Let

$$\boldsymbol{H} = \boldsymbol{N}^T \bar{\boldsymbol{Q}} \boldsymbol{N} + \bar{\boldsymbol{R}} \tag{28}$$

$$\boldsymbol{g} = \boldsymbol{N}^T \bar{\boldsymbol{Q}} \left(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k) - \bar{\boldsymbol{X}}_{ref}\right) \tag{29}$$

The optimization problem underlying the model predictive controller can be formulated in the standard form of a QP problem, as follows:

$$\min_{\boldsymbol{U}_{m,\,k}} J = \frac{1}{2} \boldsymbol{U}_{m,\,k}^T \boldsymbol{H} \boldsymbol{U}_{m,\,k} + \boldsymbol{g}^T \boldsymbol{U}_{m,\,k} \tag{30}$$

For the control input $\boldsymbol{U}_{m,\,k}$, the capability constraints of the onboard thrusters must be satisfied, as is defined in Eq. (31).

$$\begin{bmatrix} \boldsymbol{I}_{6N} \\ -\boldsymbol{I}_{6N} \end{bmatrix} \boldsymbol{U}_{m,\,k} \le \begin{bmatrix} \boldsymbol{U}_{m,\,\max} \\ \vdots \\ \boldsymbol{U}_{m,\,\max} \\ -\boldsymbol{U}_{m,\,\min} \\ \vdots \\ -\boldsymbol{U}_{m,\,\min} \end{bmatrix} \in \mathbb{R}^{12N \times 1} \tag{31}$$

where $\boldsymbol{I}_{6N}$ represents the 6 N-order identity matrix. $\boldsymbol{U}_{m,\,\max}$, $\boldsymbol{U}_{m,\,\min}$ denote the maximum thrust and torque limits of the service satellite along its three axes.

3.2. Type-1 SOC: rotated cone reformulation of the quadratic cost

In this subsection, the quadratic MPC cost is reformulated as a rotated second-order cone (Type-1 SOC) to obtain an SOCP representation of the objective function.

The optimization variable is chosen as

$$\boldsymbol{z} = \begin{bmatrix} \boldsymbol{U}_{m,\,k} \\ t \end{bmatrix} \in \mathbb{R}^{(6N+1)\times 1} \tag{32}$$

The original optimization problem expressed in Eq. (30) can be equivalently written as the following optimization problem:

$$\begin{aligned} &\min_{z} \boldsymbol{f}^T\boldsymbol{z} = [\,\boldsymbol{0} \quad 1\,]\begin{bmatrix}\boldsymbol{U}_{m,\,k}\\ t\end{bmatrix} \\ &s.t. \quad \tfrac{1}{2}\boldsymbol{U}_{m,\,k}^T\boldsymbol{H}\boldsymbol{U}_{m,\,k} + \boldsymbol{g}^T\boldsymbol{U}_{m,\,k} \le t \end{aligned} \tag{33}$$

In this reformulation, a new optimization variable $\boldsymbol{z}$ is introduced, and the original objective function is reconstructed into an SOC constraint by means of an auxiliary variable t.

Let $\boldsymbol{H} = \boldsymbol{L}^T\boldsymbol{L}$ (the matrix $\boldsymbol{L}$ can be obtained using Cholesky decomposition), and define

$$\begin{cases} y_1 = t - \boldsymbol{g}^T\boldsymbol{U}_{m,\,k} \\ y_2 = 1 \\ y_3 = \boldsymbol{L}\boldsymbol{U}_{m,\,k} \end{cases} \tag{34}$$

The SOC constraint in Eq. (33) is therefore equivalent to

$$\begin{aligned} &\tfrac{1}{2}\boldsymbol{U}_{m,\,k}^T\boldsymbol{H}\boldsymbol{U}_{m,\,k} + \boldsymbol{g}^T\boldsymbol{U}_{m,\,k} \le t \\ &\qquad\qquad \Updownarrow \\ &2y_1y_2 \ge \| y_3 \|_2^2, \quad y_1 \ge 0, \quad y_2 \ge 0 \end{aligned} \tag{35}$$

which corresponds to the standard definition of the rotated Lorentz cone. The above rotated cone can be equivalently represented by a standard Lorentz cone.

$$\| \boldsymbol{A}_{soc}^{(1)}\boldsymbol{z} - \boldsymbol{b}_{soc}^{(1)} \| \le \boldsymbol{d}_{soc}^{(1)\,T}\boldsymbol{z} - \boldsymbol{\gamma}_{soc}^{(1)} \tag{36}$$

i.e.,

$$\| \begin{bmatrix}\sqrt{2}\boldsymbol{L} & \boldsymbol{0}\\ -\boldsymbol{g}^T & 1\end{bmatrix}\boldsymbol{z} - \begin{bmatrix}0\\1\end{bmatrix} \| \le [\,-\boldsymbol{g}^T \quad 1\,]\boldsymbol{z} - (-1) \tag{37}$$

This rotated cone corresponds to the Type-1 SOC constraint in the overall SOCP–MPC formulation, and the coefficient matrices of each term in Eq. (37) can be described as:

$$\boldsymbol{A}_{soc}^{(1)} = \begin{bmatrix}\sqrt{2}\boldsymbol{L} & \boldsymbol{0}_{6N\times 1}\\ -\boldsymbol{g}^T & 1\end{bmatrix}, \quad \boldsymbol{b}_{soc}^{(1)} = \begin{bmatrix}\boldsymbol{0}_{6N\times 1}\\ 1\end{bmatrix},$$
$$\boldsymbol{d}_{soc}^{(1)} = [\,-\boldsymbol{g}^T \quad 1\,], \quad \boldsymbol{\gamma}_{soc}^{(1)} = -1.$$

3.3. Type-2 SOC: flight corridor constraint

During the proximity phase, the service satellite should follow the flight corridor. The flight corridor can typically be represented by an SOC, as illustrated in Fig. 3. This subsection introduces the Type-2 SOC constraint that encodes the flight corridor geometry as a conic set.

The cone is defined by its axis direction and half-angle $\phi_{\max}$ constraint. In this study, the corridor axis is chosen to align with the target's body-fixed X-axis.

$$\hat{\boldsymbol{e}}_{tbx}^{tb} = [\,1 \quad 0 \quad 0\,]^T \tag{38}$$

At the i-th prediction step within the receding horizon, the projection of the corridor axis direction onto the reference orbital frame can be expressed as $\hat{\boldsymbol{e}}_{tbx,\,i}^{o}$.

The flight corridor constraint at the i-th predicted moment can be expressed as an SOC constraint

$$\angle\left(\boldsymbol{r}_{mt,\,i}, \hat{\boldsymbol{e}}_{tbx,\,i}^{o}\right) \le \phi_{\max} \tag{39}$$

Eq. (39) states that the angle between the line-of-sight vector from the target to the service satellite and the corridor axis is smaller than the half-angle constraint $\phi_{\max}$ of the SOC, indicating that the service satellite remains within the flight corridor.

By substituting Eq. (21) into Eq. (39), it can be further expressed as

$$\begin{aligned} &\left\| \begin{array}{l} \left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}_{tbx,\,i}^{o}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\right)\boldsymbol{E}_{r,i}\boldsymbol{N}\boldsymbol{U}_{m,\,k} + \\ \left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}_{tbx,\,i}^{o}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\right)\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)) \end{array} \right\| \\ &\le \tan\phi_{\max}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\boldsymbol{E}_{r,i}\boldsymbol{N}\boldsymbol{U}_{m,\,k} \\ &+ \tan\phi_{\max}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)) \end{aligned} \tag{40}$$

where $\boldsymbol{E}_{r,\,i} = [\,\boldsymbol{I}_3 \quad \boldsymbol{0}_{3\times 9}\,]$.

Eq. (40) defines the Type-2 SOC constraint in the SOCP problem, which ensures that the service satellite remains inside the prescribed flight corridor, and the corresponding coefficient matrices can be expressed as:

$$\boldsymbol{A}_{soc,i}^{(2)} = \left[\,\left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}_{tbx,\,i}^{o}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\right)\boldsymbol{E}_{r,i}\boldsymbol{N} \quad \boldsymbol{0}_{3\times 1}\,\right],$$
$$\boldsymbol{b}_{soc,i}^{(2)} = -\left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}_{tbx,\,i}^{o}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\right)\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)),$$
$$\boldsymbol{d}_{soc,i}^{(2)} = \left[\,\tan\phi_{\max}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\boldsymbol{E}_{r,i}\boldsymbol{N} \quad 0\,\right]^T,$$
$$\boldsymbol{\gamma}_{soc,i}^{(2)} = -\tan\phi_{\max}\hat{\boldsymbol{e}}_{tbx,\,i}^{oT}\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)).$$

3.4. Type-3 SOC: pointing constraint

In addition, during the proximity phase, it is required that the target remain within an SOC (see Fig. 3) whose axis is aligned with the X-axis of the service satellite body frame and whose half-angle is $\delta_{\max}$. This Type-3 SOC constraint describes the pointing requirement, ensuring that the target lies inside a cone centered around the payload boresight.

The X-axis of the service satellite body frame can be expressed as

$$\hat{\boldsymbol{e}}_{mbx}^{mb} = [\,1 \quad 0 \quad 0\,]^T \tag{41}$$

At the i-th prediction step within the receding horizon, the projection of the pointing SOC axis direction onto the reference orbital frame can be expressed as $\hat{\boldsymbol{e}}_{mbx,\,i}^{o}$.

The pointing constraint at the i-th predicted moment can be expressed as an SOC constraint

$$\angle\left(\boldsymbol{r}_{mt,i}, \hat{\boldsymbol{e}}^{o}_{mbx,\,i}\right) \le \delta_{\max} \tag{42}$$

Eq. (42) states that the angle between the vector from the service satellite to the target and the payload direction (cone axis) is smaller than $\delta_{\max}$, indicating that the target lies within the admissible capture region.

Eq. (42) can be further expressed as

$$\begin{aligned}&\left\| \begin{array}{l} \left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}^{o}_{mbx,\,i}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\right)\boldsymbol{E}_{r,i}\boldsymbol{N}\boldsymbol{U}_{m,\,k} + \\ \left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}^{o}_{mbx,\,i}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\right)\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)) \end{array} \right\| \\ &\le \tan\delta_{\max}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\boldsymbol{E}_{r,i}\boldsymbol{N}\boldsymbol{U}_{m,\,k} \\ &+ \tan\delta_{\max}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)) \end{aligned} \tag{43}$$

Eq. (43) defines the Type-3 SOC constraint used in the SOCP formulation, and the corresponding coefficient matrices can be expressed as:

$$\begin{aligned}
\boldsymbol{A}^{(3)}_{soc,i} &= \left[\left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}^{o}_{mbx,\,i}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\right)\boldsymbol{E}_{r,i}\boldsymbol{N} \quad \boldsymbol{0}_{3\times 1}\right], \\
\boldsymbol{b}^{(3)}_{soc,i} &= -\left(\boldsymbol{I}_3 - \hat{\boldsymbol{e}}^{o}_{mbx,\,i}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\right)\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)), \\
\boldsymbol{d}^{(3)}_{soc,i} &= \left[\tan\delta_{\max}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\boldsymbol{E}_{r,i}\boldsymbol{N} \quad 0\right]^T, \\
\boldsymbol{\gamma}^{(3)}_{soc,i} &= -\tan\delta_{\max}\hat{\boldsymbol{e}}^{oT}_{mbx,\,i}\boldsymbol{E}_{r,i}(\boldsymbol{M}\boldsymbol{X}(k) + \boldsymbol{\Gamma}\boldsymbol{U}_t(k)).
\end{aligned}$$

3.5. Time-varying cone tightening and SOCP–MPC algorithm

This subsection introduces a time-varying cone tightening strategy for the Type-2 and Type-3 SOC constraints and summarizes the overall SOCP–MPC algorithm.

Eqs. (37), (40), and (43) together constitute all SOC constraints. The final form of the reformulated SOCP-based optimization problem can be expressed as

$$\begin{aligned}
&\min_{\boldsymbol{z}} \boldsymbol{f}^T\boldsymbol{z} = [\boldsymbol{0} \quad 1]\begin{bmatrix}\boldsymbol{U}_{m,\,k} \\ t\end{bmatrix} \\
&s.t.\begin{cases} \boldsymbol{A}_{ineq}\boldsymbol{z} \le \boldsymbol{b}_{ineq} \\ \| \boldsymbol{A}_{soc(j)}\boldsymbol{z} - \boldsymbol{b}_{soc(j)} \| \le \boldsymbol{d}^T_{soc(j)}\boldsymbol{z} - \boldsymbol{\gamma}_{soc(j)}, \; j = 1, \cdots, n_{cones}. \end{cases}
\end{aligned} \tag{44}$$

where the number of SOC $n_{cones} = 1 + 2(N+1)$.

For the cone half-angle limits $\phi_{\max}$ and $\delta_{\max}$ in the Type-2 flight corridor SOC constraint and the Type-3 pointing SOC constraint, we innovatively propose a dynamic time-varying SOC constraint method.

As the service satellite approaches the target from its initial position and enters the flight corridor (see Fig. 4), the cone half-angle is gradually tightened. For example, when the satellite is at the position at time 1, the current cone half-angle is ϕ_1.

To satisfy the SOCP solving conditions and ensure recursive feasibility and constraint satisfaction during optimization, the half-angle constraint is set to $\phi_{\max,\,1}$. During the approach, the same process applies to time 2, and so forth, until the satellite enters the desired cone half-angle range ϕ_{des} at time 3.

Throughout the process, the half-angle constraint satisfies the following condition:

$$\phi_{\max} = \begin{cases} k_{fcc}\phi, & \phi \ge \phi_{\text{des}} \\ \phi_{\text{des}}, & \phi < \phi_{\text{des}} \end{cases} \tag{45}$$

During the approach phase, the dynamic tightening of the pointing constraint follows a similar principle. The service satellite continuously performs attitude maneuvers, adjusting the cone axis direction while progressively compressing the cone half-angle, as illustrated in Fig. 5.

Accordingly, the half-angle bound of the Type-3 pointing SOC constraint is formulated as

$$\delta_{\max} = \begin{cases} k_{pc}\delta, & \delta \ge \delta_{\text{des}} \\ \delta_{\text{des}}, & \delta < \delta_{\text{des}} \end{cases} \tag{46}$$

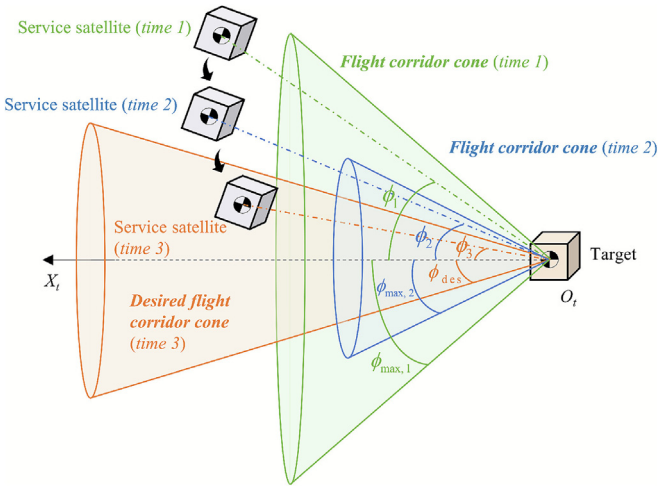


Fig. 4. Time-varying flight corridor cones.

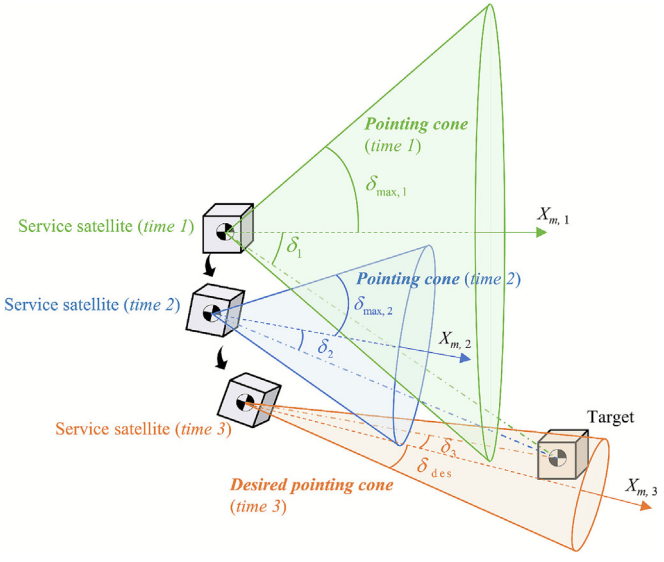


Fig. 5. Time-varying pointing cones.

Table 1
Pseudo code of SOCP–MPC algorithm.

Algorithm 1 SOCP–MPC Algorithm

% **Input**
Sampling time Δt, prediction horizon N
Weights: Q, R
Actuator limits: $\boldsymbol{U}_{m,\,\max}$, $\boldsymbol{U}_{m,\,\min}$, $\boldsymbol{F}_{t,\,\max}$
Mission params: desired corridor half-angle ϕ_{des} and factor k_{fcc}, desired pointing half-angle δ_{des} and factor k_{pc}
% **State/Control**
State $\boldsymbol{X} = \begin{bmatrix} \boldsymbol{r}_{mt}^T & \boldsymbol{v}_{mt}^T & \boldsymbol{q}_{mt}^T & \boldsymbol{\omega}_{mt}^T \end{bmatrix}^T$
Control $\boldsymbol{U}_m = \begin{bmatrix} \boldsymbol{F}_m^T & \boldsymbol{T}_m^T \end{bmatrix}^T$
% **Procedure**
for *t = 0: timestep: simulation end time*
1: Update state & geometry: initial state $\boldsymbol{X}(k)$, reference state $\boldsymbol{X}_{ref}(k)$, corridor half-angle ϕ, pointing half-angle δ.
2: Update 6-DOF discrete model $\boldsymbol{A}_d$, $\boldsymbol{B}_{d,\,m}$, $\boldsymbol{B}_{d,\,t}$ and calculate predictive model matrices $\boldsymbol{M}$, $\boldsymbol{N}$, $\boldsymbol{\Gamma}$ % Eq. (21)
3: Calculate original optimization problem matrices $\boldsymbol{H}$, $\boldsymbol{g}$ % Eqs. (28), (29)
4: Calculate SOCP optimization problem matrices $\boldsymbol{L}$, $\boldsymbol{A}_{soc,\,i}^{(1)}$, $\boldsymbol{b}_{soc,\,i}^{(1)}$, $\boldsymbol{d}_{soc,\,i}^{(1)}$, $\gamma_{soc,\,i}^{(1)}$ Eqs. (36), (37)
5: Update the cone half-angle limit % Eq. (45) and calculate flight corridor SOC constraints params $\boldsymbol{A}_{soc,\,i}^{(2)}$, $\boldsymbol{b}_{soc,\,i}^{(2)}$, $\boldsymbol{d}_{soc,\,i}^{(2)}$, $\gamma_{soc,\,i}^{(2)}$ % Eq. (40)
6: Update the cone half-angle limit (Eq. (46).) and calculate pointing SOC constraints params $\boldsymbol{A}_{soc,\,i}^{(3)}$, $\boldsymbol{b}_{soc,\,i}^{(3)}$, $\boldsymbol{d}_{soc,\,i}^{(3)}$, $\gamma_{soc,\,i}^{(3)}$ % Eq. (43)
7: Call SOCP solver within the allotted time and extract the output at the current moment $\boldsymbol{U}_m$.
8: Apply $\boldsymbol{U}_m$ to the system % Eq. (11)
end for

By introducing scaling factors k_{fcc} and k_{pc}, the cone constraints are dynamically tightened, thereby accelerating the service satellite's entry into the desired flight corridor and pointing with the target, while simultaneously ensuring the solvability of the optimization problem.

In this section, the main procedure of the proposed SOCP–MPC control algorithm with dynamic tightening constraints is summarized in Algorithm 1.

4. Numerical simulations

4.1. Mission scenario and configurations

In the OPE rendezvous and pointing phase, the mission objective is defined as guiding the service satellite to reach the reference position with a relative position error less than 0.01 m and a relative velocity below 0.01 m/s. The process is required to be rapid, fuel-efficient, and to satisfy the desired flight corridor and pointing conditions as early as possible. Let the desired flight corridor half-angle be denoted as $\phi_{\text{des}} = 20^{\circ}$ and the desired pointing cone half-angle as $\delta_{\text{des}} = 10^{\circ}$.

In this section, to validate the effectiveness of the proposed SOCP–MPC control method through simulation, the target is assumed to possess residual maneuvering capability and to operate under a worst-case full three-axis thrusting strategy, as detailed in Appendix A. This setup serves as a rational opponent in the simulations, providing a representative testing background for evaluating the service satellite's control method. In addition, the baseline control strategy of the service satellite, along with the corresponding control parameters, is presented in Appendix B. The baselines are implemented using a control scheme based on differential games combined with PD feedforward attitude control, denoted as DG–PD. In one baseline, no reference trajectory guidance is included, the desired state is directly set to the mission objective, denoted as DG–PD–NG. In the other baseline, an intermediate reference waypoint is manually prescribed to facilitate rapid corridor entry, denoted as DG–PD–G. The remaining detailed simulation setups and results are presented as follows.

Both the service satellite and the target operate in GEO. The service satellite is initially located at [3500; 600; −3000] m relative to the target. As a preparatory step for ADR transfer, the desired position of the service satellite is set to [50; 0; 0] m, with its payload direction (body minus X-axis) aligned toward the target's CM.

The mass properties and maneuvering capabilities of the service satellite and the target are summarized in Table 2. The service satellite and the target start with zero angular velocity. The initial quaternion vector part of the service satellite is $\boldsymbol{q}_{m0} = [0.1 \quad 0.1 \quad 0]^T$, while that of the target is set to zero.

For evaluating control effort consumption, the comparison metric is chosen as the propellant mass consumed during the OPE process, denoted as m_{fuel}. Its expression is given by:

$$m_{fuel} = \int_0^{t_f} \frac{|F_{mx}| + |F_{my}| + |F_{mz}|}{I_{sp} g_0} dt \tag{47}$$

In the expression, the gravitational acceleration g_0 is taken as 9.80665 m/s^2.

The control parameters employed in the SOCP–MPC scheme designed in Section 3 are specified as follows:

Table 2
Configuration parameters of the service satellite and the target.

Service satellite parameters	Value	Target parameters	Value
Mass/kg	2000	Mass/kg	3500
Moment of inertia/kgṡm^2	3000	Moment of inertia/kgṡm^2	4500
Maximum control thrust/N	100	Maximum control thrust/N	50
Specific impulse I_{sp}/s	300	–	–
Maximum control torque/Nṡm	5	–	–

$$\boldsymbol{Q} = \boldsymbol{F} = \begin{bmatrix} 0.2 \cdot \boldsymbol{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \begin{bmatrix} 200 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{bmatrix} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 8 \cdot \boldsymbol{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 20 \cdot \boldsymbol{I}_3 \end{bmatrix},$$

$$\boldsymbol{R}_m = \begin{bmatrix} 10^{-8} \cdot \boldsymbol{I}_3 & \mathbf{0} \\ \mathbf{0} & 10^{-8} \cdot \boldsymbol{I}_3 \end{bmatrix}, \quad \boldsymbol{R}_t = \begin{bmatrix} 10^{-8} \cdot \boldsymbol{I}_3 & \mathbf{0} \\ \mathbf{0} & 10^{-8} \cdot \boldsymbol{I}_3 \end{bmatrix}.$$

4.2. Performance comparison with DG–PD baselines

Based on the designed control strategy and the selected parameters, numerical simulations were conducted to validate the maneuvering process of a GEO service satellite, which approaches the target along the flight corridor path and performs attitude alignment toward the target.

Fig. 6 illustrates the OPE trajectories of the service satellite and the target in the orbital coordinate frame under the SOCP–MPC method. The trajectories of DG–PD–NG and DG–PD–G exhibit similar patterns and are therefore not shown. Nevertheless, it can be observed that all three cases achieve a smooth approach.

Figs. 7–9 compare the thrust component variations of the service satellite under the three control schemes. With the SOCP–MPC method (Fig. 7), the thrust components converge more rapidly to low amplitudes, although small corrective thrusts are required in the later stage to counter the target's maneuvers. In contrast, the two DG–PD baseline schemes (Figs. 8 and 9) maintain high thrust levels over a longer duration, resulting in greater propellant expenditure. As shown in the cumulative fuel consumption in Fig. 10, the SOCP–MPC method achieves significantly lower propellant usage throughout the maneuver, demonstrating superior efficiency.

Figs. 11–13 illustrate the relative velocity between the service satellite and the target. In the results of SOCP–MPC and DG–PD–NG, the service satellite and the target generally follow a strategy of accelerating first and then decelerating. However, in the DG–PD–G case, the manual switching of the guidance point leads to an alternating "accelerate–decelerate–accelerate–decelerate" pattern, which induces additional propellant consumption.

Figs. 14–16 present the torque variations of the service satellite under different control methods. In the initial phase, all three methods exhibit a rapid response, followed by a swift decay toward near-zero values. For the two DG–PD baseline methods, the output torques are almost identical due to attitude-orbit decoupling and the use of identical attitude control parameters. In contrast, under the SOCP–MPC method, small periodic oscillations of constant amplitude appear in the later phase, caused by the target's repeated maneuvers.

Fig. 17 shows the relative trajectory from the perspective of the target body frame, while Fig. 18 provides a magnified view of the terminal approach. The cone with the solid line marking its axis indicates the desired flight corridor

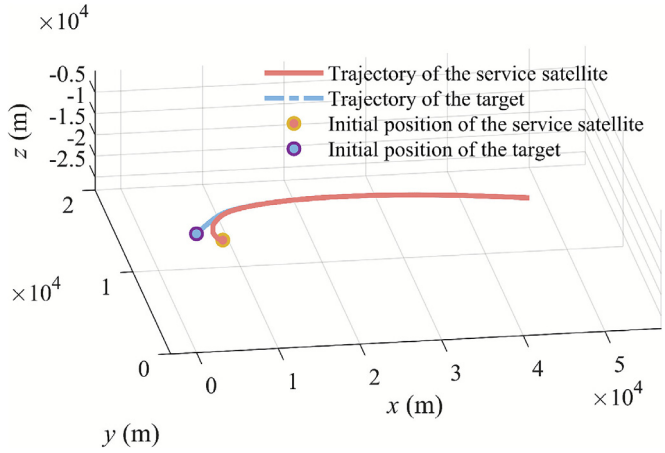


Fig. 6. Trajectory of the service satellite and the target (SOCP–MPC).

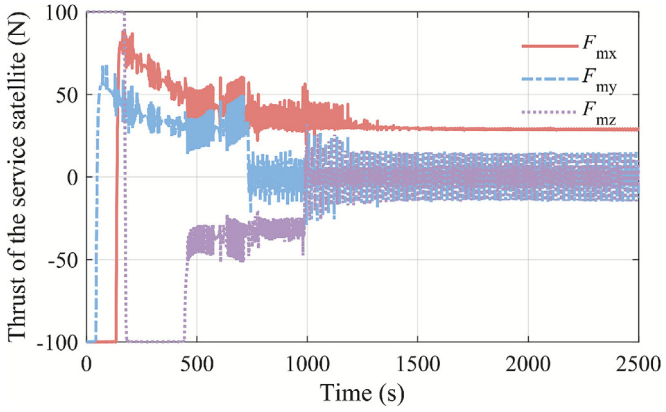


Fig. 7. Thrust of the service satellite (SOCP–MPC).

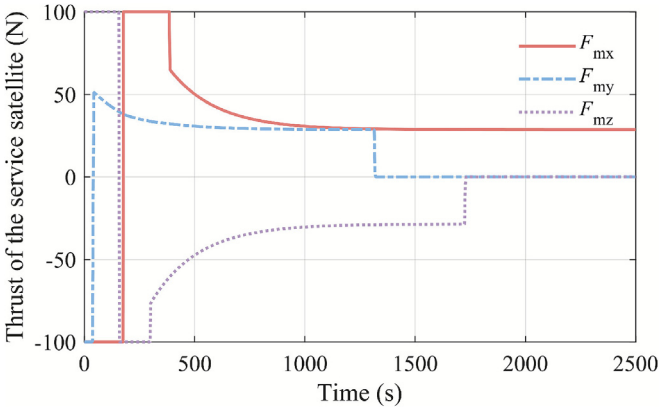


Fig. 8. Thrust of the service satellite (DG–PD–NG).

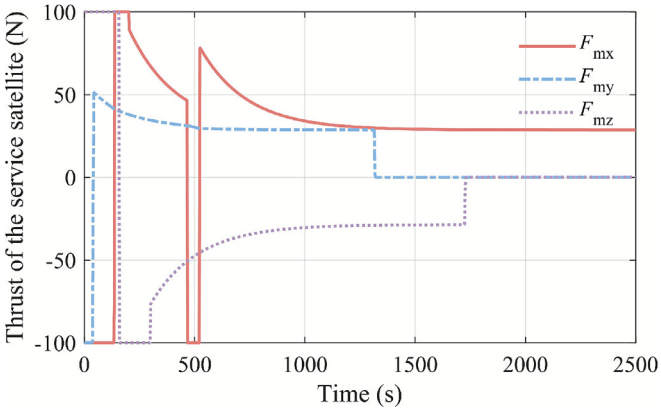


Fig. 9. Thrust of the service satellite (DG–PD–G).

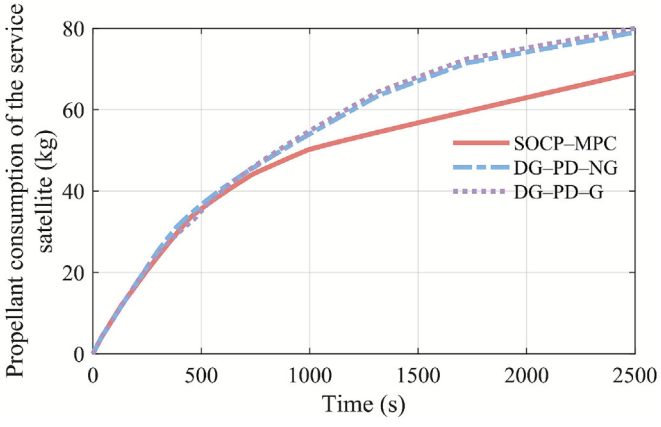


Fig. 10. Propellant consumption comparison.

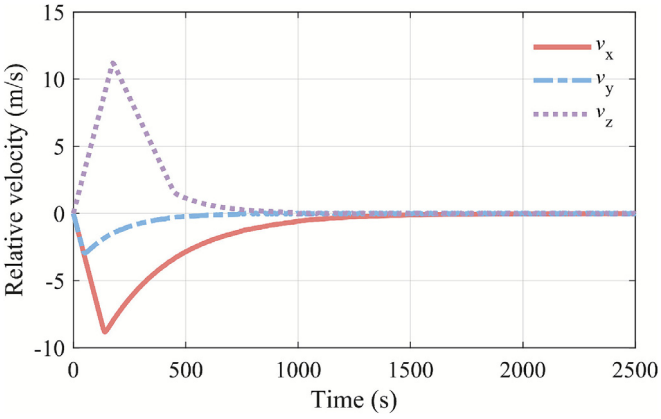


Fig. 11. Relative velocity (SOCP–MPC).

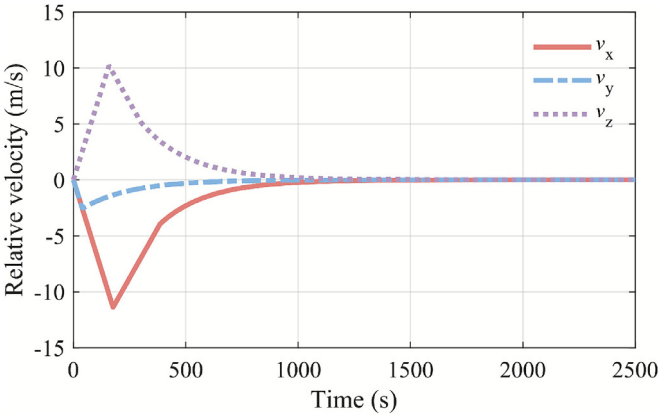


Fig. 12. Relative velocity (DG–PD–NG).

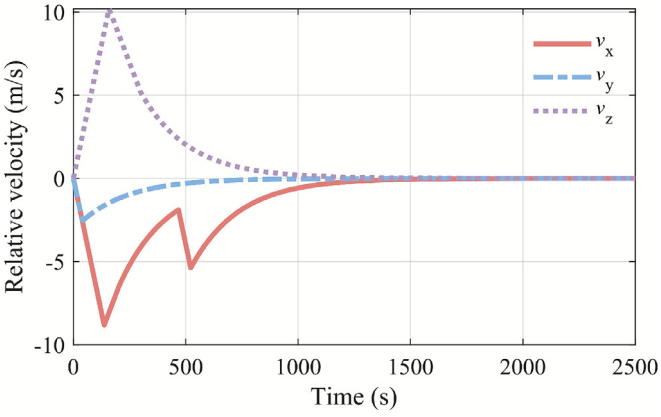


Fig. 13. Relative velocity (DG–PD–G).

with a half-angle of 20 degrees. Other smaller cones represent the desired pointing cone with a half-angle of 10 degrees, overlapping at the initial time.

From the analysis of Figs. 17–18 regarding flight corridor entry, the SOCP–MPC method enables entry from the farthest relative distance, with the DG–PD–G case performing almost comparably. The trajectories then converge toward the final desired position along the vicinity of the cone axis. Within tens to hundreds of meters from the target, the SOCP–MPC trajectory remains more closely aligned with the cone axis. In contrast, the DG–PD–NG method stays outside the corridor for most of the time, only cutting in near the terminal phase, with a significant lateral offset remaining upon entry.

In terms of attitude pointing, the SOCP–MPC trajectory benefits from coupled orbit–attitude predictive control, enabling faster attitude adjustments and smoother transi-

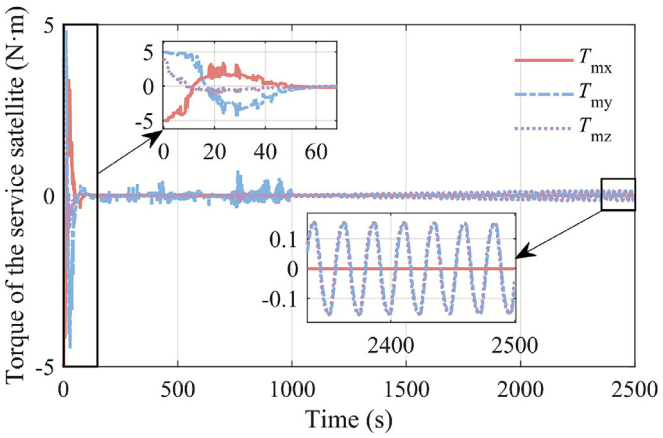


Fig. 14. Torque of the service satellite (SOCP–MPC).

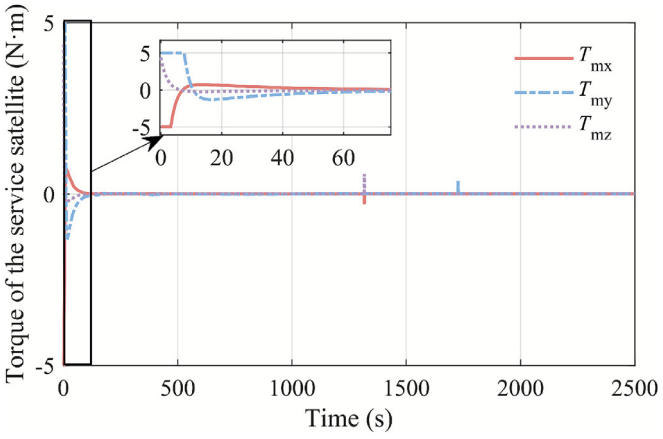


Fig. 15. Torque of the service satellite (DG–PD–NG).

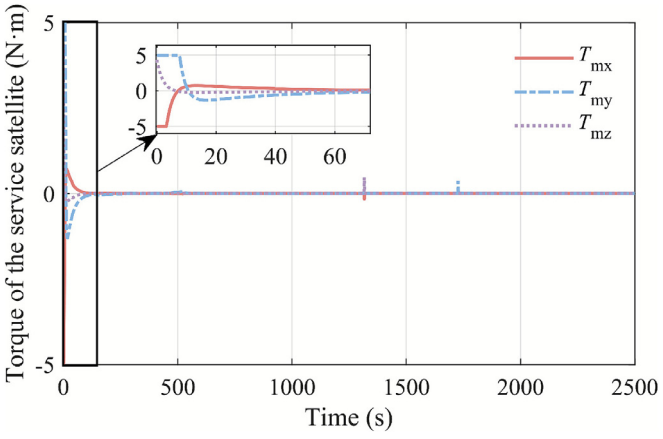


Fig. 16. Torque of the service satellite (DG–PD–G).

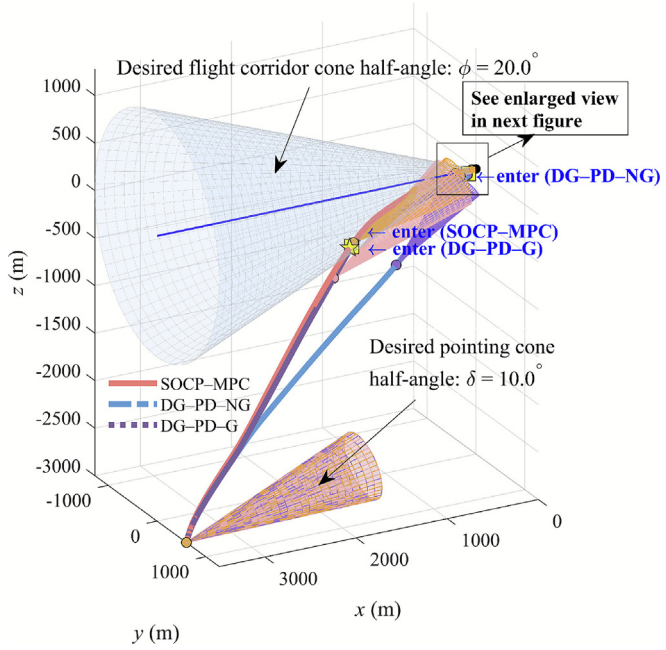


Fig. 17. Relative trajectory comparison (from the view of the target).

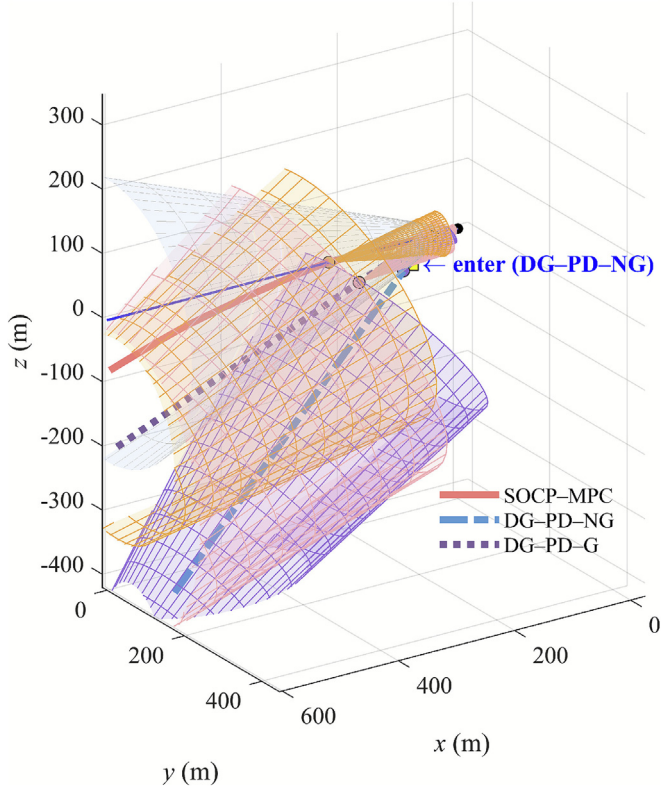


Fig. 18. Enlarged view of Fig. 17 near the terminal approach.

tions. As a result, the pointing cone encompasses the target earlier and maintains this condition for a longer duration, thereby providing favorable observation conditions and creating a stable window for payload deployment. Both DG–PD baselines exhibit noticeable delays in achieving proper alignment.

Fig. 19 presents the projected relative trajectories under the three methods. The horizontal axis denotes the distance along the flight corridor cone axis, while the vertical axis represents the shortest perpendicular distance from each trajectory point to the cone axis, as defined in Fig. 18.

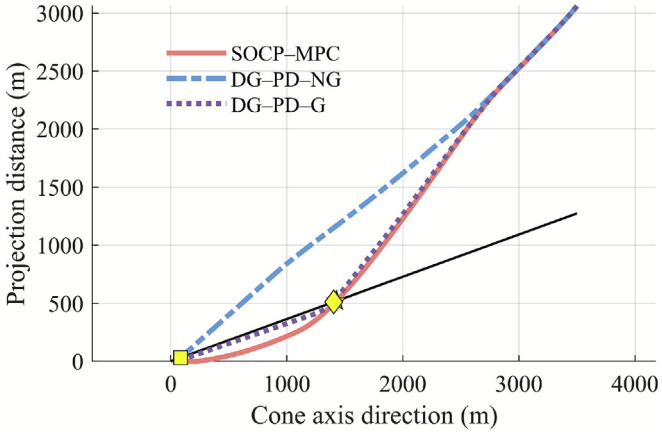


Fig. 19. Relative trajectory projection comparison.

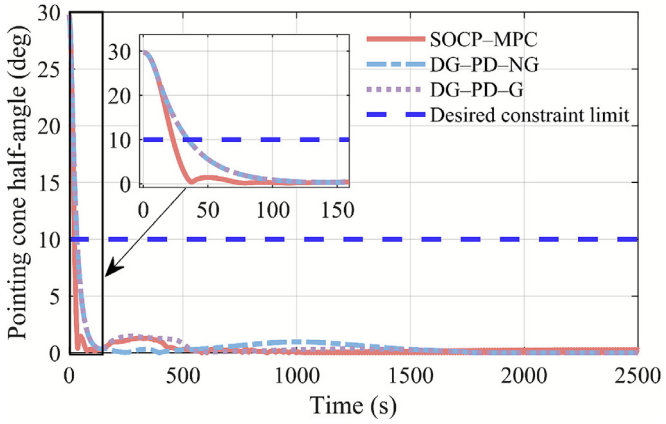


Fig. 21. Pointing cone half-angle comparison.

The solid straight line indicates the desired corridor boundary. The three markers denote the corridor entry points. SOCP–MPC (1428.96 m, 520.2 m) and DG–PD–G (1404.3 m, 511.3 m) enter at nearly the same relative position. Along the axial direction, all three methods remain within the boundary once entering the corridor. The advantage of SOCP–MPC lies in its lateral accuracy, as the lateral projection at the same axial distance consistently remains smaller than that of the DG–PD methods.

Fig. 20 illustrates the angle between the vector from the target to the service satellite and the flight corridor axis, along with the desired cone half-angle constraint of 20 degrees. The SOCP–MPC method enters the desired flight corridor at 364.5 s, while DG–PD–G takes slightly longer at 467.6 s. The DG–PD–NG baseline exhibits a significantly delayed entry, requiring 1059.5 s, and converges at a slower rate. In terms of convergence speed, SOCP–MPC demonstrates a clear advantage, reaching near-zero angular deviation more than 600 s earlier than the baselines.

Fig. 21 shows the angle between the payload pointing direction and the target, with the desired cone half-angle constraint of 10 degrees. All three methods achieve rapid attenuation and maintain small angular errors over an extended duration, with SOCP–MPC performing slightly faster than the two DG–PD baselines.

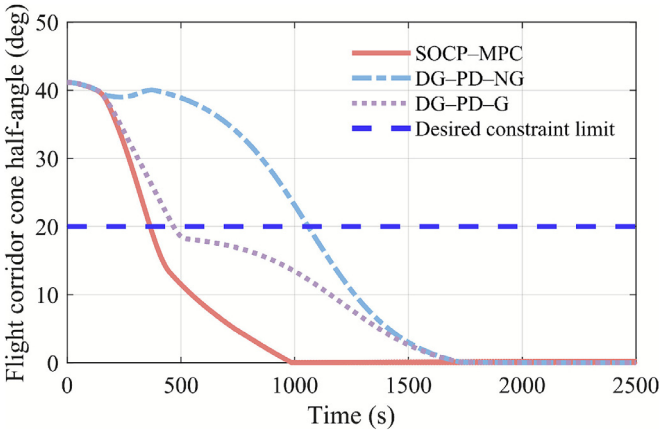


Fig. 20. Flight corridor cone half-angle comparison.

In the OPE rendezvous and pointing task, the primary performance metrics of interest include the total time required to reach the desired position, the propellant consumption, the entry time into the desired flight corridor, and the time to satisfy the pointing requirement. The first three rows of Table 3 summarize the key performance indicators of the nominal SOCP–MPC method compared with the DG–PD baselines.

For DG–PD–NG, since the final position is directly set as the objective, the flight distance is relatively the shortest. However, it enters the desired flight corridor the latest (1059.5 s). Although the overall mission time is the shortest (1770.0 s), the trajectory involves repeated rapid accelerations and decelerations, leading to substantial propellant expenditure (71.9 kg). By contrast, SOCP–MPC and DG–PD–G enter the corridor at nearly the same relative position, and their approach trajectories are generally similar, resulting in comparable mission durations (1967.6 s vs. 1974.0 s). Nevertheless, as indicated by the relative velocity profiles (Figs. 11 and 13), SOCP–MPC achieves globally planned, smooth maneuvers with efficient propellant utilization (62.5 kg vs. 74.9 kg). In DG–PD, the manually introduced intermediate guidance point inevitably causes additional deceleration and re-acceleration, which delays corridor entry (467.6 s) compared with SOCP–MPC (364.5 s).

Regarding pointing, SOCP–MPC employs a unified optimization of thrust and torque, coordinating attitude and orbital control. Consequently, it achieves pointing alignment slightly earlier (21.8 s) than DG–PD–G (34.6 s).

Overall, earlier entry into the flight corridor generally comes at the expense of a longer total mission duration. In comparison, the SOCP–MPC method autonomously guides the service satellite into the flight corridor without relying on manually designed reference waypoints, achieving faster entry while substantially reducing propellant consumption without significantly sacrificing total mission time.

Table 3
Key performance metrics.

Metric/Unit	Time to complete the mission/s	Propellant consumption until mission completion/kg	Flight corridor entry time/s	Pointing alignment start time/s
SOCP–MPC	1967.6	62.5	364.5	21.8
DG–PD–NG	1770.0	71.9	1059.5	34.6
DG–PD–G	1974.0	74.9	467.6	34.6
SOCP–MPC–Fast	1573.1	60.5	455.6	22.9
SOCP–MPC–Slow	2318.3	65.6	339.9	23.1

In addition to the performance metrics, the computational burden of the SOCP–MPC scheme was evaluated. For a 2500 s simulation with a control sampling period of 0.1 s (25 000 control updates), all SOCP problems were solved on a desktop computer equipped with an Intel i7-13700KF CPU. The average CPU time per control step was about 0.05 s using a standard SOCP solver, which is approximately one half of the sampling period. Although this hardware is more powerful than a typical onboard processor, these results indicate a reasonable margin for real-time implementation, especially when combined with tailored embedded implementation and code optimization.

4.3. Parameter study and trade-off analysis

For the SOCP–MPC method, the corridor entry timing can be flexibly adjusted through the control parameters, rather than depending on manually prescribed guidance points as in DG–PD. The previously discussed case is denoted as SOCP–MPC–Nominal. By tuning the control parameters in Eq. (22),

$$\boldsymbol{Q}=\boldsymbol{F}=\begin{bmatrix} 0.2\cdot\boldsymbol{I}_3 & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \begin{bmatrix} 130 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{bmatrix} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & 8\cdot\boldsymbol{I}_3 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & 20\cdot\boldsymbol{I}_3 \end{bmatrix}.$$

A faster case scenario, denoted as SOCP–MPC–Fast, is obtained.

Tune the control parameters as follows:

$$\boldsymbol{Q}=\boldsymbol{F}=\begin{bmatrix} 0.2\cdot\boldsymbol{I}_3 & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \begin{bmatrix} 260 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{bmatrix} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & 8\cdot\boldsymbol{I}_3 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & 20\cdot\boldsymbol{I}_3 \end{bmatrix}.$$

A slower case scenario, denoted as SOCP–MPC–Slow, is obtained.

Figs. 22 and 23 present the relative trajectories and their projections under different control parameter settings. Combined with the performance comparison in Table 3, it can be observed that tuning the SOCP–MPC control parameters enables flexible adjustment of mission performance metrics such as the corridor entry time. Moreover, the entry time into the flight corridor exhibits a clear inverse relationship with the total time required to complete the approach task.

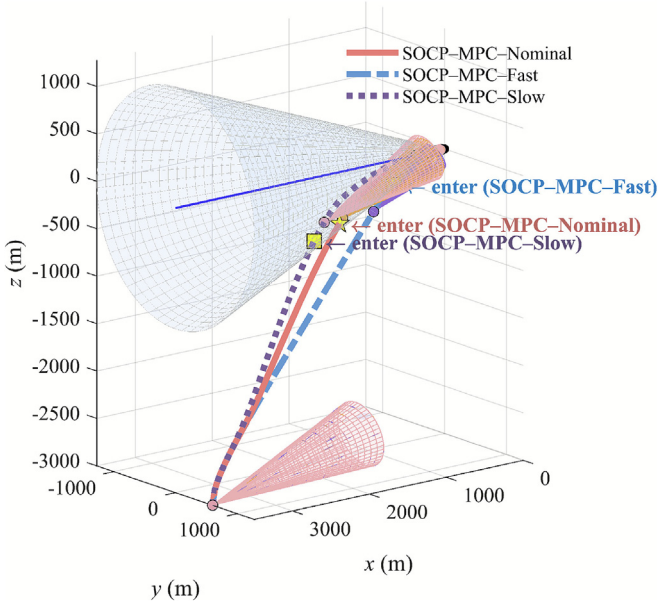


Fig. 22. Relative trajectory comparison (SOCP–MPCs).

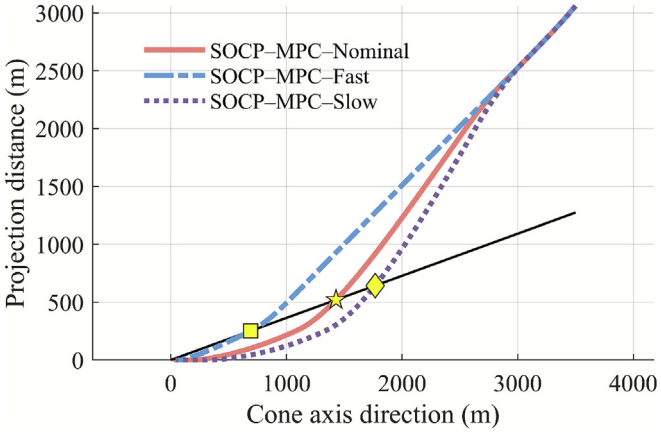


Fig. 23. Relative trajectory projection comparison (SOCP–MPCs).

A degree of robustness is inherent to the proposed framework, which combines receding-horizon optimization with time-varying cone tightening. Provided the actual target maneuvers remain within the assumed thrust bounds, modeling errors primarily lead to more conservative trajectories and elevated propellant consumption rather than a loss of feasibility.

In summary, the proposed SOCP–MPC framework provides a practically implementable and systematically superior control scheme for the OPE rendezvous and pointing phase of ADR missions. By dynamically tightening SOC constraints within a convex optimization framework, the method autonomously guides the service satellite to the desired position, prioritizes the satisfaction of flight corridor and pointing constraints during the maneuver, and reduces propellant consumption.

5. Conclusions

This study presents a unified SOCP–MPC framework for the OPE rendezvous and pointing phase in ADR missions targeting defunct satellites. The quadratic MPC cost is reformulated as a rotated SOC, while the flight corridor and pointing requirements are encoded as time-varying SOC constraints within an orbit–attitude coupled model. A dynamic cone-tightening strategy is incorporated to account for persistent target maneuvers while preserving convexity on the service satellite side. The resulting conic program can be solved onboard in real time using standard conic optimization solvers. Numerical simulations indicate earlier corridor entry, faster pointing alignment, and lower propellant consumption compared with DG–PD baselines, as well as tunable trade-offs between mission duration and fuel usage.

The main contributions are:

(1) A unified SOCP–MPC formulation for autonomous OPE rendezvous and pointing in ADR missions integrating orbit–attitude dynamics and actuation limits into a single convex optimization framework.
(2) A control-oriented SOC formulation of the quadratic MPC cost and two mission constraints, leading to a convex conic optimization problem.
(3) A dynamic cone-tightening strategy is incorporated to maintain convexity, thereby enabling a constraint-driven approach and pointing without reliance on pre-planned trajectories.
(4) A numerical assessment, including comparisons with DG–PD baselines and a parameter study, demonstrating earlier corridor entry, faster pointing convergence, reduced propellant usage, and a tunable trade-off between mission duration and fuel consumption.

Future work will focus on improving terminal accuracy and robustness, in particular by explicitly incorporating target maneuver uncertainty and model errors into the SOCP–MPC design, and on extending the framework to tethered-tugging control and more complex ADR scenarios.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Target evasion control policy

This evasive policy is used only to test the proposed controller and the baseline controller. The target remains three axis stabilized (no attitude maneuvers) and performs translation only thrusting to increase the relative separation from the service satellite.

The thrust component F_t expressed in the body-fixed frame are determined by the relative position error x_{mt} along the corresponding axis and follow the prescribed dead-zone and sign logic, as formulated below.

$$F_t = \begin{cases} F_{t,\max}, & x_{mt} < -x_{deadzone} \\ 0, & -x_{deadzone} \le x_{mt} \le x_{deadzone} \\ -F_{t,\max}, & x_{mt} > x_{deadzone} \end{cases} \tag{A1}$$

where $x_{deadzone}$ denotes the dead-zone threshold, set to 0.1 m in the simulation tests, and $F_{t,\max}$ denotes the maximum thrust along a single axis of the target.

Appendix B. Baseline DG–PD control strategies

B.1. DG OPE algorithm

The relative orbital dynamics of the service satellite and the target are described in Eq. (1). In line with the DG formulations in Tang et al. (2021) and Ye et al. (2021), the cost function of a two-player zero-sum game is defined as

$$J_{dg} = \frac{1}{2}\int_0^{t_f} \Big((\boldsymbol{x}_{mt} - \boldsymbol{x}_{ref})^T \boldsymbol{Q}_{dg} (\boldsymbol{x}_{mt} - \boldsymbol{x}_{ref}) + \boldsymbol{U}_{dg,\,m}^T \boldsymbol{R}_{dg,\,m} \boldsymbol{U}_{dg,\,m} - \boldsymbol{U}_{dg,\,t}^T \boldsymbol{R}_{dg,\,t} \boldsymbol{U}_{dg,\,t} \Big) dt \tag{B1}$$

where $\boldsymbol{x}_{ref}$ denotes the reference state (including the position and the velocity); $\boldsymbol{Q}_{dg}$, $\boldsymbol{R}_{dg,\,m}$, $\boldsymbol{R}_{dg,\,t}$ are control weights; $\boldsymbol{U}_{dg,\,m}$, $\boldsymbol{U}_{dg,\,t}$ denotes the control outputs of the service satellite and the target.

For baseline without manual guidance DG–PD–NG, the reference state is as follows:

$$\boldsymbol{x}_{ref} = [\,50 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0\,]^T \tag{B2}$$

For baseline with manual guidance DG–PD–G, the reference state is as follows:

$$\boldsymbol{x}_{ref} = \begin{cases} [1000 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T, & \phi > \phi_{\text{des}} \\ [50 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T, & \phi \le \phi_{\text{des}} \end{cases} \tag{B3}$$

The Hamilton-Jacobi-Bellman (HJB) framework is used to solve the Riccati equation below (Tang et al., 2021; Ye et al., 2021).

$$\begin{aligned} &\boldsymbol{A}_{pos}^T \boldsymbol{P}_{dg} + \boldsymbol{P}_{dg} \boldsymbol{A}_{pos} \boldsymbol{P}_{dg} \left(\boldsymbol{B}_{pos,\,m} \boldsymbol{R}_{dg,\,m}^{-1} \boldsymbol{B}_{pos,\,m}^T \right. \\ &\left. - \boldsymbol{B}_{pos,\,t} \boldsymbol{R}_{dg,\,t}^{-1} \boldsymbol{B}_{pos,\,t}^T / (u_{m,\,\max}/u_{t,\,\max})^2 \right) \boldsymbol{P}_{dg} \\ &+ \boldsymbol{Q}_{dg} = 0 \end{aligned} \tag{B4}$$

where $\boldsymbol{P}_{dg}$ is the control matrix; $u_{m,\,\max}$ and $u_{t,\,\max}$ denote the maximum acceleration along a single axis of the service satellite and the target.

The optimal closed-loop feedback control law can thus be obtained.

$$\boldsymbol{U}_{dg,\,m}^* = -F_{m,\,\max} \text{sat} \left(\frac{\boldsymbol{R}_{dg,\,m}^{-1} \boldsymbol{B}_{dg,\,m} \boldsymbol{P}_{dg} \boldsymbol{x}_{mt}}{F_{m,\,\max}} \right) \tag{B5}$$

where $\text{sat}(\cdot)$ denotes the saturation function, $F_{m,\,\max}$ denotes the maximum thrust along a single axis of the service satellite.

In the simulation tests, the control weights are set as

$$\boldsymbol{Q}_{dg} = \begin{bmatrix} 4.45 \times 10^{-1} \cdot \boldsymbol{I}_3 & \boldsymbol{0} \\ \boldsymbol{0} & 2.05 \times 10^4 \cdot \boldsymbol{I}_3 \end{bmatrix},$$

$$\boldsymbol{R}_{dg,\,m} = 1 \times 10^{-3} \cdot \boldsymbol{I}_3, \quad \boldsymbol{R}_{dg,\,t} = 1 \times 10^{-3} \cdot \boldsymbol{I}_3.$$

B.2. PD feedforward attitude control law

Following Schlanbusch et al. (2012), we designed a simple attitude-pointing PD feedforward controller. First, we specify the quaternion $\boldsymbol{Q}_p$, angular velocity $\boldsymbol{\omega}_p$, and angular acceleration $\dot{\boldsymbol{\omega}}_p$ of the service satellite pointing frame with respect to the orbital frame. Then, given the current quaternion $\boldsymbol{Q}_m$ of the service satellite, we compute the error quaternion $\boldsymbol{Q}_{mp} = \begin{bmatrix} q_{mp0} & \boldsymbol{q}_{mp}^T \end{bmatrix}^T$ and the error angular velocity $\boldsymbol{\omega}_{mp}$. Finally, the control law is obtained as

$$\begin{aligned} \boldsymbol{T}_{mc} = &-k_p \text{sgn}(q_{mp0}) \boldsymbol{J}_m \boldsymbol{q}_{mp} - k_d \boldsymbol{J}_m \boldsymbol{\omega}_{mp} + \boldsymbol{\omega}_m^{\times} \boldsymbol{J}_m \boldsymbol{\omega}_m \\ &- \boldsymbol{J}_m \boldsymbol{\omega}_{mp}^{\times} \boldsymbol{C}_{mp} \boldsymbol{\omega}_p + \boldsymbol{J}_m \boldsymbol{C}_{mp} \dot{\boldsymbol{\omega}}_p \end{aligned} \tag{B6}$$

where k_p and k_d are the proportional and derivative control gains, and $\boldsymbol{C}_{mp}$ denotes the coordinate transformation matrix from the pointing frame to the service satellite body frame.

In the simulation tests, the control gains are set as $k_p = 0.035$ and $k_d = 0.5$.

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